08 — Testing and Evaluation Plan

Prepared 9 November 2026 for teacher review on 10 November. **Draft; not approved or implemented.**

Planning-stage submission: test specifications are provided; actual results are Not run. Evidence of results belongs after execution. The teacher capacity is 50–150; numeric quality thresholds are proposals awaiting agreement. Never substitute simulated SFU load for physical Wi-Fi evidence.

Source: 01-testing-strategy.md

Testing strategy

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1. Objectives and evidence status

Verify source requirements, permissions, local runtime, media behavior and safe policy recovery. All product tests in this package are **Not run**; they are specifications for later execution. Documentation checks are reported separately in the validation report.

2. Test levels

Level	Focus	Example evidence
Business-rule tests	Validation, transitions, idempotency, seat allocation	Concurrent last-seat approval; stale operation rejected
Component integration	API/database/SFU/adapter boundaries	Scoped grants, accepted-versus-verified policy result
End-to-end	Teacher/student/admin workflows	F01/F05/F06/F07/F08/F10/F13 outcomes
Fault/recovery	Isolated component failures and restart	Durable operations, truthful unresolved states, safe reconciliation
Performance	Declared physical/simulated loads and media quality	Raw timestamp samples, client metrics, server/AP topology
Security	Role/scope/media and secret boundaries	Denied cross-class calls, revoked subscriber, secret inspection
User acceptance	Teacher task completion and required results	Signed or pending disposition per requirement

3. Entry criteria

Approved test scope and network authority; isolated/safe fixtures; declared build, configuration and browser/OS; known baseline and recovery plan; agreed acceptance targets or explicit exploratory label. Use synthetic identities and test domains; do not collect real student browsing history as a test shortcut.

4. Exit criteria

Execute all agreed required tests; link evidence to requirements/stories; triage defects and rerun affected scenarios after changes; report optional omissions and scope gaps. Release blockers follow Definition of Done. Teacher acceptance is separately recorded, with any exceptions and their authority.

5. Execution and defect record

Test ID, requirement, story, build, config, environment, tester, timestamp, preconditions, steps, expected result, observed result, pass/fail/blocked, evidence, defect ID, reviewer and retest reference. An unreachable dependency is a recorded blocked/failure outcome, never automatic success. Criteria approval and execution outcome occupy separate fields.

Source: 02-test-cases.md

Test-case catalogue

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1. Catalogue authority and execution convention

The 63 criteria below are preserved from the existing requirements acceptance catalogue. They remain **Proposed**; results are **Not run**. Each case needs the detailed environment and evidence record from strategy. Steps are specified as the executable scenarios within each criterion; before running, instantiate named devices, class, domains, credentials and expected values. Numeric conditions refer to AT proposals, not approved commitments.

2. Original requirement cases

Requirement	Story	Test ID	Procedure and expected result
NAC01	US-NAC01	TC-NAC01	In a controlled authorized scope, enable and disable a profile; status and client-path probes show the effective result. Disabled/unreachable integration is not reported as verified.
NAC02	US-NAC02	TC-NAC02	For each configured category, predeclare representative denied domains and allowed controls; application produces the expected DNS/gateway results. Record list version and known coverage gaps.
NAC03	US-NAC03	TC-NAC03	A custom blocked domain is denied and an allowed domain is permitted through the configured enforcement path; invalid domain syntax is rejected before a network change.
NAC04	US-NAC04	TC-NAC04	Run allow-by-default and restricted-by-default profiles against the same declared test set; results match the agreed rule/exception precedence, including subdomain handling under D07.

Requirement	Story	Test ID	Procedure and expected result
NAC05	US-NAC05	TC-NAC05	Starting an authorized class with an approved profile creates an apply operation and audit outcome; healthy-scope verification meets proposed AT04. Starting from another class/scope cannot apply it.
NAC06	US-NAC06	TC-NAC06	After applying a temporary profile, end the class; prior effective rules and allowed/blocked client-path controls are restored within proposed AT05 when reachable. Restoration failures remain visible and retryable.
NAC07	US-NAC07	TC-NAC07	Save two named profiles, reload the application/backend and retrieve equivalent rule versions; switching the selected profile does not itself falsely report network enforcement.
NAC08	US-NAC08	TC-NAC08	Given a blocked category, add an approved domain exception; that control resolves/connects where upstream permits, while another category domain remains denied. Protect local app/media access throughout.
NAC09	US-NAC09	TC-NAC09	Schedule a start/end against a declared timezone and test clock; each executes once per agreed idempotency semantics after restart, rechecks scope approval and records actual result. Missed-run policy awaits D11.
NAC10	US-NAC10	TC-NAC10	Compare desired version, adapter read-back and affected-client probe results; UI accurately differentiates saved, applying, verified, degraded, failed and restoration-pending outcomes with timestamps.
NAC11	US-NAC11	TC-NAC11	Perform a profile edit, apply, failed apply and restore; each audit record includes actor/service identity, action, scope, time and outcome, with no credentials.
NAC12	US-NAC12	TC-NAC12	Apply restrictions then invoke authorized emergency restore; prior policy verifies under proposed AT05. Unauthorized request changes no rules; an unreachable adapter yields a visible unresolved result.
NAC13	US-NAC13	TC-NAC13	Apply a rule with declared expiry; close the admin browser and restart the backend before expiry. Recovery executes restoration once safely and verifies it under AT05 or displays pending/failure if unreachable.

Requirement	Story	Test ID	Procedure and expected result
NAC14	US-NAC14	TC-NAC14	Export and reimport a sample profile; normalized rule content matches, secrets are absent, malformed imports are rejected and applying requires its own authorized operation.
NAC15	US-NAC15	TC-NAC15	Where the chosen adapter supports isolated identity/scope, apply a group rule to target A and demonstrate that control B retains its prior policy. Otherwise mark unsupported and record the scope decision.
NAC16	US-NAC16	TC-NAC16	Execute the role-permission matrix's representative allowed and denied operations; Student/Teacher roles alone cannot edit policies or infrastructure, and out-of-scope administrators cannot mutate that scope.
NAC17	US-NAC17	TC-NAC17	Inspect built frontend, network responses, profile exports and logs for test secret markers; none appear. Adapter secrets/signing keys are accessible only through protected server configuration.
NAC18	US-NAC18	TC-NAC18	Apply the most restrictive agreed profile; a fresh student join, teacher controls, live media, local names and restoration remain functional. Reject an explicit conflicting local-service rule.
NAC19	US-NAC19	TC-NAC19	Review app/resolver logging and storage using synthetic browsing events; collect only agreed minimal data, mask/exclude unnecessary raw histories and demonstrate deletion according to D13.
NAC20	US-NAC20	TC-NAC20	After provisioning local lists, disable WAN and restart services; profiles/rules remain retrievable and local test-domain policy operations work without remote list fetching. Baseline offline behavior remains required.
T01	US-T01	TC-T01	Create 10 concurrent active classes with the same authorized teacher or test teachers; all session IDs and active join codes are distinct. Closed-code join is rejected.
T02	US-T02	TC-T02	Run start → pause → resume → stop on the supported setup; both viewers see each agreed state and publication stops after stop. Stopping sharing leaves the class open; pause/audio semantics await D08.
T03	US-T03	TC-T03	For every supported capture type (screen/window/tab), publish the teacher-selected source to two viewers; denial/cancellation creates no published capture. Record unsupported browser/source combinations under D09.

Requirement	Story	Test ID	Procedure and expected result
T04	US-T04	TC-T04	On each agreed OS/browser/source combination, verify shared audio when supported, microphone audio when selected, and no microphone publication when not selected. Record unsupported sources explicitly.
T05	US-T05	TC-T05	Join two viewers and disconnect one; roster shows the correct participant count and each current connection state within proposed 5 s after the server detects the event.
T06	US-T06	TC-T06	Demonstrate approve, reject, remove and block separately. Removed access closes; blocked rejoin fails for the agreed identity/duration. Repeat cross-class control attempts and verify denial; D02/D03 define guarantees.
T07	US-T07	TC-T07	Apply two supported quality profiles; record configured and observed resolution, frame rate and bitrate with adaptation limits. Unsupported input is rejected or reported; do not assert exact network bitrate from settings alone.
T08	US-T08	TC-T08	From a second laptop, open the displayed URL and decoded QR destination; both point to the intended local class/join route and the displayed code joins that class.
T09	US-T09	TC-T09	During a controlled media fault, diagnostics distinguish connection state, missing/stalled media and available WebRTC statistics; display measurement time and refresh within proposed 5 s.
T10	US-T10	TC-T10	Exercise one approved presentation, PDF, image and video sample through the agreed sharing method; admitted viewers can access it and rejected formats show a useful error. D16 defines upload versus media presentation behavior.
T11	US-T11	TC-T11	End two owned classes; retained history shows their correct IDs/times and participant counts. Another teacher cannot read them; summaries do not imply video recordings.
S01	US-S01	TC-S01	From two test laptops on the selected network, the local URL and QR destination both reach the join page with trusted browser access. Record page-ready time against proposed AT01.
S02	US-S02	TC-S02	In both proposed modes, missing, invalid, expired or closed codes are rejected. Code admission permits viewing only after final checks; code + approval keeps a valid request pending until owning-Teacher approval and fresh atomic admission/capacity checks. Wrong guesses follow the agreed rate-limit rule; execute TC-ADM01–TC-ADM07.
S03	US-S03	TC-S03	Two admitted viewers receive the selected live screen via the local SFU; non-admitted clients cannot subscribe. Measure physical-viewer playback and target-scale conditions under AT02/AT03.
S04	US-S04	TC-S04	On supported audio combinations, each viewer can start playback and adjust/mute its own audio independently; another viewer’s volume and teacher publication are unchanged.

Requirement	Story	Test ID	Procedure and expected result
S05	US-S05	TC-S05	Enter/exit fullscreen on each supported browser using user interaction; playback continues or any unsupported behavior is explained and recorded.
S06	US-S06	TC-S06	Interrupt one viewer’s network for 10 s; it reconnects to the same active class within proposed 15 s of reachability returning in at least 9/10 attempts. Removed/ended sessions never auto-readmit.
S07	US-S07	TC-S07	Inject invalid code, full class, denied admission, ended session, network loss and media failure; each produces the correct actionable status without exposing secrets.
S08	US-S08	TC-S08	With the feature enabled, enter a valid name/assigned ID; invalid input receives validation feedback and the allowed identifier is associated with the admitted participant. Identity strength and retention await D03/D13.
S09	US-S09	TC-S09	An admitted student downloads an allowed sample matching its recorded checksum; download denied when permission is off or admission is revoked; direct URL requests enforce the same checks.
S10	US-S10	TC-S10	For each selected interaction (hand/question/poll), an admitted student action reaches the owning teacher; unauthorized or closed-activity submissions are rejected. Specific interaction set awaits D17.
CN01	US-CN01	TC-CN01	Campus IT approves server name/address; two campus laptops reach the correct server with trusted access. Preserve approval and configuration evidence.
CN02	US-CN02	TC-CN02	From two representative student network locations, verify app/API and media-signaling connectivity; report each separate endpoint and any inaccessible segment.
CN03	US-CN03	TC-CN03	Establish real audio/video over the approved configured media path; record actual ports/transports and approved firewall evidence. Signaling-only success does not pass.
CN04	US-CN04	TC-CN04	Test the intended SSID/VLAN/isolation settings; students reach only the required server/enforcement paths, and guest restrictions affecting the flow are recorded.
CN05	US-CN05	TC-CN05	Record written campus IT authorization for hosting and any required firewall changes before making those changes; absence leaves this criterion pending, not passed.
CN06	US-CN06	TC-CN06	Execute staged 50/100/150 subscriber tests and AT02/AT03 on the documented campus setup; identify physical versus simulated clients and whether 150-device Wi-Fi capacity is actually demonstrated.
CN07	US-CN07	TC-CN07	Join from a campus laptop by both manual URL and QR destination; network instructions identify the correct campus network and server.

Requirement	Story	Test ID	Procedure and expected result
CN08	US-CN08	TC-CN08	Campus IT checks reserved/static server addressing against DHCP/IP allocations; conflict checks show no duplicate address before the pilot.
CN09	US-CN09	TC-CN09	With images/assets/certificates provisioned, disconnect WAN and use a fresh browser session; class creation, joining and local media work without CDN/cloud requests.
CN10	US-CN10	TC-CN10	Within written approved scope, apply an approved profile through the campus DNS/gateway adapter, verify client-path effects, and restore previous rules. Missing integration remains an acceptance gap.
OR01	US-OR01	TC-OR01	Record Wi-Fi 6-or-better equipment model/capabilities and staged load results for the intended client density; brochure client count alone is insufficient.
OR02	US-OR02	TC-OR02	Configure the intended classroom SSID/password; two student laptops associate and obtain local connectivity without changing per-class infrastructure rules.
OR03	US-OR03	TC-OR03	Inspect negotiated Wi-Fi security on the supported clients: WPA2-AES or compatible WPA3 as agreed; record any incompatible devices and resolution.
OR04	US-OR04	TC-OR04	DHCP pool/lease configuration supports at least 150 students plus inventoried support devices; test lease assignment and no duplicate allocations using declared physical/simulated conditions.
OR05	US-OR05	TC-OR05	Reboot router/server; the configured stable server address/local name still leads students to the intended application and media endpoint.
OR06	US-OR06	TC-OR06	Use and record a negotiated Gigabit-or-faster wired server link where possible; if unavailable, record the exception and measured capacity impact for acceptance.
OR07	US-OR07	TC-OR07	With intended isolation enabled/configured, two student clients still reach the server, media and approved enforcement paths; document permitted connectivity.
OR08	US-OR08	TC-OR08	Test increasing client loads, measure bottlenecks and choose AP count/placement that meets agreed AT02/AT03; repeat after relevant configuration changes.
OR09	US-OR09	TC-OR09	Demonstrate adding the supported AP/switch arrangement without address conflicts or loss of class reachability; preserve a topology/configuration record.
OR10	US-OR10	TC-OR10	Inventory required power supplies, cables/adapters and transport protection; perform one full setup and power/restart recovery using the portable kit.

Requirement	Story	Test ID	Procedure and expected result
OR11	US-OR11	TC-OR11	With WAN physically disconnected and local dependencies provisioned, start a fresh class and receive screen/supported audio on two laptops; apply/restore rules against controlled local test domains.
OR12	US-OR12	TC-OR12	Apply and restore two class profiles through the supported software adapter with no manual router rule edits between classes; unsupported automation is visibly reported as a gap.

3. Admission edge cases

Admission test	Proposed pass/fail check	References
TC-ADM01	Code admission: valid active code plus available capacity admits; invalid/expired/closed code is rejected	S02, TC-S02, F05
TC-ADM02	Code + approval: valid code creates pending request with no viewing grant; owning Teacher approval admits only after final checks	T06, TC-T06, F05
TC-ADM03	Owning Teacher rejection prevents viewing; pending/cancelled/timed-out requests do not obtain media access	T06, TC-T06; timeout policy pending D02
TC-ADM04	Student, unrelated Teacher, or unscoped administrator cannot approve a request; denied attempt does not change its pending state	NAC16, TC-AUTH, F05/F08
TC-ADM05	While approval waits, end class / expire code / add block / fill capacity in separate trials; subsequent approval does not bypass final eligibility checks	T06, S02, TC-AUTH
TC-ADM06	Two simultaneous requests compete for the final seat; at most one receives admission/media access under the agreed capacity rule	Classroom limits, S02, TC-S02
TC-ADM07	A missing join code is rejected in either mode; selecting approval mode does not create a code-free endpoint	S02, TC-S02

4. Architectural cases

Criterion/test	Proposed acceptance
AC-AUTH / TC-AUTH	Matrix allow/deny and cross-class/scope tests pass; stale/forged/revoked room grants fail; role removal affects new privileged requests; no permission is inferred only from visible UI
AC-LOCAL / TC-LOCAL	AT07 confirms local app, API, signaling, SFU, policy storage and required filtering service operation without cloud/CDN runtime dependence
AC-SFU / TC-SFU	Documented transport path shows teacher publication to local SFU and per-viewer subscriptions; actual negotiated streams/bitrates and load are recorded
AC-RECOVERY / TC-RECOVERY	Execute isolated scenarios TC-FAIL01–TC-FAIL08; health distinguishes network, application, media, trust/name and enforcement faults; unfinished temporary policy operations resume safely; backup/database/config restoration drill documented
AC-PRIVACY / TC-PRIVACY	Server secrets absent from clients/exports/logs; retained participant/audit data matches approved minimal collection and deletion rules

5. Isolated failure scenarios

Offline means WAN unavailable while the required local network, server/services, local naming and trusted access remain operational. It does not promise uninterrupted broadcasting when Wi-Fi, the server or the SFU fails. Isolate each fault below on an authorized test setup; keep the other dependencies available where the topology permits. If services share a host/process/resolver, document the coupled failure rather than calling it an independent test.

Before every trial, record build/configuration, network mode, topology, physical/simulated client counts, browser/OS, capture source, active profile, current admission and enforcement states. Capture the healthy baseline, fault start, detected outcome, recovery action, retest and final policy state. Restore the test environment before the next case. Repeat applicable cases for both campus and portable modes; campus faults require the authorized operator/test scope.

Scenario / test	Controlled fault and retained dependencies	Expected behavior during fault	Recovery and observable final state	Requirement / workflow references
FS01 / TC-FAIL01 — WAN disconnected	Disconnect upstream internet only; LAN, application/backend, SFU, local names/trust and required filtering service stay available	Fresh local joining and ongoing screen/supported-audio viewing work under AT07; public websites may fail independently of policy; no cloud runtime dependency is required	Restore WAN; local class remains usable. Use local controlled domains for offline rule checks; use upstream-connected controls separately to prove selective public-domain enforcement	CN09, OR11, NAC18/NAC20; F02/F03/F06/F10/F15
FS02 / TC-FAIL02 — One student loses Wi-Fi	Disconnect one viewer for 10 s while teacher, other viewers and local services remain reachable	Affected viewer shows lost/reconnecting status; it receives no new media during the outage; other viewers continue. Network policy does not change merely because one viewer disconnects	Restore Wi-Fi and revalidate admission; AT06 recovery applies only to a still-authorized active class. Also revoke/end while offline and verify no readmission	S06/S07, T05/T06; F08/F12
FS03 / TC-FAIL03 — Classroom LAN/AP path fails	Interrupt the class access-point/LAN path in the controlled scope; server may still be powered	Clients on that path cannot be promised broadcast, new joining or live backend status. No successful new policy verification is claimed from stale data; infrastructure recovery is required	Restore path; recheck names/trust, app, media, admission and current policy before reporting recovered. Restoration/verification stays pending until its required paths are reachable	CN02–CN04, OR07, S07; F02/F03/F12/F15
FS04 / TC-FAIL04 — Application/backend unavailable	Stop only application/backend service where separable; leave LAN/SFU/enforcement running	New joins, class controls and fresh authorization fail visibly where a client is still loaded. Existing media may continue or stop according to tested SFU/token behavior; do not promise either. Previously applied rules may remain; unreachable control plane must not be described as restored	Authorized operator restores backend; recover durable sessions/leases, reconcile ended/revoked admission and expired policies, and retest. Document any access-revocation/expiry delay; D11 must settle it before release	NAC06/NAC10/NAC13, S07; F12–F15, TC-RECOVERY
FS05 / TC-FAIL05 — SFU unavailable	Stop/disconnect only media service; application/backend and network controls remain available	App can show class/participant state while media publication/subscription fails. Display media unavailable rather than global service success. Do not reset network rules solely because the SFU failed	Restore SFU, reauthorize/resubscribe current participants and republish only with valid Teacher permission/capture consent where needed. Retain class identity where safely recoverable; report any unsupported recovery	T02–T05, S03/S04/S07; F05/F06/F12/F15

Scenario / test	Controlled fault and retained dependencies	Expected behavior during fault	Recovery and observable final state	Requirement / workflow references
FS06 / TC-FAIL06 — Local name resolution fails	Make required local hostname resolution unavailable without bypassing security; note if DNS filtering shares the fault	Fresh clients may fail before the page loads; existing cached connections may behave differently. Operator instructions identify naming failure; do not assume the app can display its own error page when unreachable	Authorized operator repairs local DNS/name configuration; verify the original trusted hostname, signaling/media and enforcement paths. A raw IP is not a passing substitute unless its secure access was explicitly provisioned	CN01/CN02, OR05, secure-context obligation; F02/F03/F15
FS07 / TC-FAIL07 — HTTPS trust/certificate fails	On an isolated test endpoint/client, present expired, untrusted or hostname-mismatched certificate; keep network reachable	Browser may block the app/WSS/capture. Do not tell users to bypass certificate checks and count that as success; use browser/endpoint evidence when app UI cannot load	Repair certificate, hostname, clock or trust provisioning as appropriate; fresh clients access original intended endpoints without warnings and can capture/connect. Do not rely on online renewal during an offline class	Secure-context obligation, CN01/CN09, OR05/OR11; F02/F03/F06/F15; D10
FS08 / TC-FAIL08 — Enforcement API or filtering service unavailable	Test adapter management-API loss separately from resolver/gateway failure where possible; app/SFU retained	Report last verified time and current unverified/degraded/pending state. Management-API loss may leave old rules effective; resolver failure may also break naming/public access. Neither proves policy disabled/restored. Preserve local access according to tested topology; expose dependencies honestly	Restore service/path; read back rules, reconcile outstanding apply/expiry/restore tasks and verify affected-client controls. Ending the class still closes app/media access when those services are reachable; network restoration may remain unresolved	NAC01/NAC06/NAC10/NAC12/NAC13/NAC18, CN10/OR12; F10/F13/F14/F15

Fault classification: FS01 tests the required offline capability. FS02–FS08 test failure detection, truthful state and recovery; they do not introduce a guarantee of uninterrupted playback through failed infrastructure. Numeric detection/recovery limits remain proposed AT targets and D11/D12 decisions. All eight results = **Not run**; requirement approval and client acceptance remain separate.

Additional cross-cutting acceptance: AC-FAILURES / TC-FAIL01–TC-FAIL08 requires the isolated fault, expected visible/operational state, recovery evidence and final policy/admission state to be recorded. Use existing CC-002/CC-017 for setup, CC-009 for viewer recovery and CC-020 for integrated fault/recovery checks; add component evidence to CC-005/CC-008/CC-012/CC-015 as relevant. These tests supplement the 63 original requirement test specifications rather than replacing or renumbering them.

6. Supplemental workflow race cases — proposed

Case	Steps / expected result	Trace
TC-CTRL01	Start F07 action, then start F13 before SFU reply; late result must not reactivate publication	US-T02; F07/F13
TC-CTRL02	Repeat same operation ID/intent after completion; return recorded result without second mutation; different intent with same ID is denied	US-T02/US-T06; F07/F08
TC-CTRL03	Resume after browser source ends; show new capture required and use F06 explicit consent	US-T02/US-T03; F07/F06
TC-CTRL04	Remove participant while SFU is unreachable; revoke app grants, record media cleanup unresolved and schedule F15	US-T06; F08/F15
TC-CLOSE01	Close media successfully while network restore fails; result is Class ended, restoration pending/failed	US-NAC06; F13
TC-CLOSE02	Fail media closure while network restore succeeds; result remains Class ending, cleanup unresolved	US-NAC06; F13
TC-CLOSE03	Repeat completed close; return original outcome; incomplete close resumes only unfinished steps using same operation	US-NAC06; F13

7. Execution template

Test	Build/environment	Criteria approval	Actual result	Evidence/defect	Reviewer/date
To select	To record	Pending	Not run	None	Pending

Source: 03-performance-testing.md

Performance and offline testing

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1. Source capacity and test conditions

The teacher specifies 50–150 students per classroom and CN06 calls for capacity validation for 150. D12 concerns acceptance test conditions and evidence. Lower-load trials cannot replace testing the source target. Record physical clients and simulated subscribers separately.

2. Test protocol proposal

1. Record server CPU/RAM/NIC/storage/OS, AP models/count/placement, SSID/VLAN, client OS/browser, SFU/build and source type.
2. Declare media profile, actual negotiated bitrate/layers, test duration and sampling method.
3. Run two-viewer functional sanity first; stage 50/100/150 subscriber trials under source AT03; lower-load trials are exploratory, not a replacement for the source target.
4. Collect per-viewer playback checks, disconnects, freezes, loss and visible timestamp latency; collect server and network utilization.
5. Repeat with actual physical devices where available. A simulator result is server-load evidence only for its declared workload.
6. Save raw samples and failure logs; stop escalating load if unsafe or unstable, investigate and record the limit.

3. Candidate metrics

Test basis	Proposed setup/target	Approval/evidence rule
AT01 — Local UI	Fresh browser reaches join page within 5 s on a reachable healthy test LAN; 10 trials per supported browser	Timing starts at navigation and ends when code input is interactive; record cold/cache conditions
AT02 — Playback quality	Agreed starting profile 720p/15 fps; source-to-display p95 ≤1 s; report received bitrate, freezes, packet loss and per-client results	Proposed threshold needs D12 approval; measure visible source timestamp at source/viewer in a common recording; separate screen content from encoder adaptation

Test basis	Proposed setup/target	Approval/evidence rule
AT03 — Capacity/stability	50/100/150 concurrent subscribers, 30 min each; per-viewer playback active at ≥98% of declared one-minute checks; no open release-blocking defect	Record physical/simulated counts, AP/SSID/topology, OS/browser, source, bitrate, server CPU/RAM/NIC and samples; simulation does not prove 150-laptop Wi-Fi airtime
AT04 — Policy apply	Healthy reachable authorized adapter reports verified outcome within 30 s of class-start/apply event in 10 trials	Must include client-path denied/allowed controls and local app/media access; upstream conditions recorded
AT05 — Restoration	Prior policy verified within 30 s of end/emergency/expiry trigger in 10 healthy reachable trials	In unreachable/conflicting-state tests, class still closes; durable pending/failure shown within proposed 5 s of detecting failure; safe retry/operator recovery required
AT06 — Reconnect	After a 10 s interruption, same-class playback recovers within 15 s of reachability returning in ≥9/10 attempts	Also test class ended and participant revoked while offline; those clients must not regain access
AT07 — Offline	Preprovision runtime assets/images, names/cert trust and lists; disconnect WAN and open fresh browser	External public-site failure is not evidence of selective blocking; use controlled local domains for offline rule checks

4. Interpretation

For latency report sample count, percentile calculation, synchronization/measurement method and outliers. For active-playback rate use successful checks divided by scheduled declared checks per viewer; disclose missing observations rather than silently excluding them. Show quality against each load stage, not only a team-wide average that hides failing viewers.

5. Report template

Build: ___ · Mode/topology: ___ · Target approved by: ___ · Actual devices/simulators: ___ · Duration: ___ · Profile/bitrate: ___ · Metrics/raw evidence: ___ · Bottleneck: ___ · Limits: ___ · Result/next action: ___

No performance, scale or offline pass is asserted by this document.

Source: 04-security-testing.md

Security testing

Version: 0.1 · **Prepared:** 9 November 2026 · **Status:** Proposed review draft; no teacher approval or product test execution claimed.

1. Authorization and test scope

Run tests only within the approved classroom/network test scope. Use synthetic users and secrets. Product security results are Not run; this document defines the review plan.

2. Scenario matrix

Test group	Positive control	Negative/abuse scenario	Expected result
TC-AUTH	Owning teacher controls own class	Student/SYS-only/other teacher uses same endpoint	Deny without state change or unauthorized data
TC-NAC16	Scoped administrator changes approved profile	Out-of-scope mutation or teacher arbitrary network edit	Deny and audit
TC-S03/S06	Admitted subscriber receives media	Expired/forged/revoked grants; reconnect after removal/end	No unauthorized subscription/refresh
TC-ADM05/06	Eligible pending request approved	Concurrent final seat; block/close while pending	Atomic allocation; stale request does not bypass checks
TC-NAC17	Server uses test secret	Search client bundles, responses, exports and logs for marker	No secret disclosure
TC-NAC18	Restrictive profile preserves local paths	Rule conflicts with required local names/control/media	Reject unsafe configuration or report failure safely
TC-NAC06/12/13	Restore owned changes	External IT edit, partial apply, restart, unreachable service	Reconcile, preserve unrelated changes, no false restored status
TC-NAC19	Minimal audit fields retained	Unexpected browsing histories or excessive identifiers	Remove/limit collection; match approved D13

Test group	Positive control	Negative/abuse scenario	Expected result
TC-CTRL01/02	Current control operation succeeds	Stale publication version, mismatched reused operation ID	No replay or resurrection

3. Evidence and reporting

Record endpoint/action, authenticated role, class/scope, pre-state, request, observed effect, sanitized evidence and expected denial. UI button visibility is not a security test. Separate finding severity, affected requirement, fix owner, retest and release disposition. Never publish real credentials in the report.

4. Limitations

Domain-control tests do not certify whole-device restrictions. Record bypass conditions and accepted limitations with the network owner. No compliance certification, penetration-test completion or browser support certification is implied.

Source: 05-user-acceptance-testing.md

User acceptance testing

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1. Purpose and participants

Teacher/client evaluates agreed classroom tasks and release constraints; campus IT evaluates authorized infrastructure behavior. Hour coordinates evidence. Actual reviewers, test venue and dates remain to be named. User acceptance is distinct from a Sprint Review or successful diagram rendering.

2. Acceptance session script

Task	Expected observable result	Trace
Create and start class	Correct joining information; start is authorized	UC01/F04/F01
Join valid/invalid/pending	Only admitted student gains viewing rights	UC02/F05
Broadcast screen and supported audio	Correct source delivered; limitations explicit	UC03/F06
Pause/resume/stop	Confirmed state; stop sharing leaves class open	UC03/F07
Manage participants	Correct target action; revoked media and recovery status truthful	UC04/F08
Apply profile	Saved configuration separated from verified affected-client result	UC05/06/F09/F10
Disconnect/reconnect	Revalidate current access; revoked/ended user denied	UC07/F12
End and restore	Class closure and restoration independently reported	UC08/F13
Expiry/recovery	Durable result and authorized next action	UC09/10/F14/F15
Offline and agreed capacity	Declared physical/simulated evidence meets approved criteria	AT02/03/07

3. Acceptance record

Requirement / scenario	Build and evidence	Reviewer	Accepted / rejected / pending	Exception and authority	Date
To complete			Pending		

4. Completion conditions

All agreed required criteria have reviewed evidence; release blockers resolved; omissions and limitations explicitly approved; user/operator guides and recovery handover reviewed. Optional features are accepted only if included. No signature, satisfaction score or acceptance result has been fabricated.